

Reviewed Preprint

v1 • August 26, 2025

Not revised

Reviewed Preprint

v2 • November 28, 2025

Revised by authors

✉ For correspondence:

wadih.arap@rutgers.edu

eyd10@rutgers.edu

Author notes: See [page 16](#)

Funding: See [page 15](#)

Reviewing editor: Yongliang Yang,
Shanghai University of Medicine and
Health Sciences, China

© 2025, Cadoná et al. This article is
distributed under the terms of the
[Creative Commons Attribution
License](#), which permits unrestricted
use and redistribution provided that
the original author and source are
credited.

Endoscopic liquid biopsies of gastric fluid in a large patient cohort reveal DNA content as a candidate tumor biomarker in gastric cancer

Francine Carla Cadoná^{1,2}, Thais F Bartelli^{1,*}, Adriane Graicer Pelosof³, Claudia Zitron Sztokfisz³,
Adriana Passos Bueno⁴, Luana Batista do Carmo dos Santos¹, Gabriela Pereira Branco¹,
Gabriel Oliveira dos Santos⁴, Warley Abreu Nunes⁴, Fernanda Araújo Pintor⁵, Laís Lie Senda de Abrantes⁵,
Alexandre Defelicibus⁶, Luiz Gonzaga Vaz Coelho⁷, Marcis Leja⁸, Haejin In^{9,10}, Sharon Li^{11,12},
Howard Hochster^{9,13}, Felipe José Fernandez Coimbra¹⁴, Rodrigo D Drummond⁶, Israel Tojal da Silva⁶,
Ravi J Chokshi^{11,15}, Renata Pasqualini^{11,16}, Wadih Arap^{11,17}✉, Diana Noronha Nunes^{1,11,16},
Emmanuel Dias-Neto^{1,11,16}✉

¹Laboratory of Medical Genomics, A. C. Camargo Cancer Center, São Paulo, Brazil • ²Masters in Health and Life Sciences, Franciscana University, Santa Maria, Brazil • ³Endoscopy Section, A. C. Camargo Cancer Center, São Paulo, Brazil • ⁴Department of Pathology, A. C. Camargo Cancer Center, São Paulo, Brazil • ⁵Research Support Center, A. C. Camargo Cancer Center, São Paulo, Brazil • ⁶Laboratory of Computational Biology, A. C. Camargo Cancer Center, São Paulo, Brazil • ⁷Postgraduation Program in Applied of Sciences of Adult Health & Alpha Gastroenterology Institute, Hospital das Clínicas, Faculty of Medicine, Federal University of Minas Gerais, Belo Horizonte, Brazil • ⁸University of Latvia, Institute of Clinical and Preventive Medicine and Center for Gastric Diseases GASTRO, Riga, Latvia • ⁹Rutgers Cancer Institute, New Brunswick, United States • ¹⁰Division of Surgical Oncology, Department of Surgery, Rutgers Robert Wood Johnson Medical School, New Brunswick, United States • ¹¹Rutgers Cancer Institute, New Brunswick, United States • ¹²Division of Hematology/Oncology, Department of Medicine, Rutgers New Jersey Medical School, Newark, United States • ¹³Division of Medical Oncology, Department of Medicine, Rutgers Robert Wood Johnson Medical School, New Brunswick, United States • ¹⁴Department of Abdominal Surgery, A. C. Camargo Cancer Center, São Paulo, Brazil • ¹⁵Division of Surgical Oncology, Department of Surgery, Rutgers New Jersey Medical School, Newark, United States • ¹⁶Division of Cancer Biology, Department of Radiation Oncology, Rutgers New Jersey Medical School, Newark, United States • ¹⁷Division of Hematology/Oncology, Department of Medicine, Rutgers New Jersey Medical School, Newark, United States

eLife Assessment

This **valuable** work substantially advances our understanding of prognostic value of total gfDNA in gastric cancer. The evidence supporting the conclusions is **solid**, supported by a large, well-classified patient cohort and controlled clinical variables. The work will be of broad interest to scientists and clinical pathologist working in the field of gastric cancer.

<https://doi.org/10.7554/eLife.107103.2.sa3>

Abstract

Background

Human gastric cancer remains a diagnostic and therapeutic challenge worldwide. Improved prognostic biomarkers may support treatment planning, including surgery, neoadjuvant and adjuvant therapies. We offer a new liquid biopsy approach, integrated with

esophagogastroduodenoscopy (EGD) to evaluate gastric fluid recovered from a large patient cohort (n=1,056 patients), demonstrating the value of a simple DNA concentration analysis to provide diagnostic support and prognosis in gastric cancer.

Methods

We performed an exploratory study to evaluate gastric fluid DNA (gfDNA) concentration in patients with normal gastric mucosa or diagnosed with peptic diseases, preneoplastic conditions, or cancer. Key variables including sex, gastric fluid pH, use of proton-pump inhibitors, tumor subtype, clinical stage, and disease outcomes were considered.

Results

gfDNA concentrations were significantly increased in gastric cancer versus all groups (mean 26.86 ng/μL; 95% CI: 20.05 to 33.79; $p=3.61e^{-12}$) and as compared to non-malignant controls (normal mucosa and peptic diseases; mean 10.77 ng/μL; 95% CI: 9.23 to 12.33; $p=9.55e^{-13}$), and preneoplastic conditions (mean 10.10 ng/μL; 95% CI: 7.59 to 12.60; $p=1.10e^{-5}$). gfDNA concentrations were higher in advanced tumors (T3; mean 25.66 ng/μL; 95% CI: 19.46 to 31.85) compared to early-stage disease (T2 and below; mean 15.12 ng/μL; 95% CI: 9.73 to 20.50; $p=5.97e^{-4}$). Our findings also suggest gfDNA prognostic value for gastric cancer diagnosed subjects. In this subset, patients with gfDNA concentrations >1.28 ng/μL showed longer progression-free survival ($p=0.009$), which correlated with increased tumor-infiltrating immune cells ($p=0.001$), remaining significant after adjusting for tumor stage ($p=0.014$).

Conclusions

While high gfDNA indicates gastric cancer in a general population and may contribute to disease management, its diagnostic clues should be integrated with further work up such as tissue biopsies, to ensure comprehensive and accurate clinical assessment. However, in patients with established gastric cancer, gfDNA is a potential prognostic marker, and subjects with high gfDNA content might paradoxically have better outcomes. The elevated presence of immune cell infiltrates—compared to those observed in peptic diseases and pre-neoplastic lesions (including their associated gastric cavity DNA shedding)—may serve as a critical missing link to resolve this empirical paradox.

Introduction

Esophagogastroduodenoscopy (EGD) is routinely performed to work up vague digestive complaints and is used to biopsy suspicious lesions and ultimately diagnose gastric cancer, a tumor that ranks as the fifth in terms of incidence and mortality in the world, with current estimates of ~660,000 annual deaths globally (Globocan, 2022 [\[1\]](#)). In areas of high-incidence EGD has been shown to contribute to a reduction in gastric cancer mortality (Hamashima et al., 2013 [\[2\]](#); Matsumoto and Yoshida, 2014 [\[3\]](#)), and is also employed for presurgical gastric cancer staging and even as a treatment method (endoscopic resection) in early-stage disease (Kim and Jung, 2021 [\[4\]](#)), rendering it a universally used tool in this setting. During EGD, gastric fluids are collected and discarded, allowing a better evaluation of the gastric mucosa.

The study of DNA found in body-fluid samples (e.g., blood, urine, saliva, cerebrospinal fluid, peritoneal washes) can support the diagnosis and/or follow-up of cancer (Forshew et al., 2012 [\[5\]](#)). In a liquid biopsy approach, tumor-derived DNA permits the monitoring of specific mutations diluted in the fluids under investigation. As the gastric fluid is in direct contact with the stomach mucosa and gastric cancer lesions, liquid biopsies performed with gastric fluid are neither restricted to just the few selected areas sampled by tissue biopsy, nor by the minute amounts of tumor DNA that might be diluted in peripheral blood. Also, it may indeed allow a full representation of cells that reach the stomach, derived from the upper digestive tract, including immune cells, tumor cells and the microbiota. In a prior feasibility study, we showed that gastric fluid incidentally collected from gastric cancer patients during EGD contains tumor-derived DNA,

allowing the capture of tumor mutations (Pizzi et al., 2019 [DOI](#)). We have now collected gastric fluids from a large patient cohort (n=1,056), obtained during diagnostic endoscopy, to investigate whether gastric fluid DNA (gfDNA) concentrations would vary according to disease diagnosis, including the presence of non-malignant, premalignant, or malignant lesions, gastric cancer subtype and stage, tumor location, and patient age or sex. Notably, we also evaluated whether gfDNA concentrations would have predictive and prognostic value on gastric cancer recurrence and patient outcomes.

Methods

Patients

All subjects enrolled in this study underwent diagnostic EGD at a single institution, the A. C. Camargo Cancer Center, São Paulo, Brazil. Inclusion criteria required subjects to be above 18 years old, with clinical indication for performing a diagnostic EGD examination. Samples and analyses were excluded if requested by the participants (even after the signature of a written informed consent), or if other unrelated diagnoses were found (e.g., previous gastrectomy or presence of other non-gastric cancer). Samples were collected at diagnosis, during standard EGD, from February 2016 to March 2021. EGDs were only performed after referral from an independent physician (i.e., not related to the study) to assess abdominal symptoms, under standard sedation. Samples were obtained after either patients or their guardians signed a written informed consent form approved by the Clinical Research Committee, the Ethics Research Committee, and the Institutional Review Board (IRB) (protocol #2134/15). Diagnostic and epidemiological data were collected as described (Bartelli et al., 2019 [DOI](#)).

Gastric fluid collection, DNA extraction and quantification

As usual and customary for upper endoscopic examination, all subjects were fasting for 8-12 hours. At the beginning of the procedures, gastric fluids that are routinely removed to allow better examination of the mucosa, were collected in sterile plastic containers attached to the endoscope suction channel and kept on ice until pH measurement and neutralization. Samples were divided in aliquots and kept frozen at -20°C (each aliquot was thawed just once), until DNA was extracted from an equal volume of gastric fluid (800 µl) by proteinase-K digestion, followed by phenol-chloroform extraction and ethanol DNA precipitation. DNA was subsequently resuspended in 100 µl of sterile nuclease-free double-deionized water and quantified by Qubit fluorometry (Thermo Fisher).

Quantification of human- versus bacteria-derived DNA in gastric fluids

For a representative fraction of the liquid biopsies (n=180, 19.1%), including all 10 control subjects with no clinical findings (i.e., no pathologies detected on EGD) along with peptic diseases (n=51), premalignant lesions (n=55), and gastric cancer cases (n=64), we evaluated the proportion of human- versus bacteria-derived DNA in gastric fluids by using quantitative PCR (Albuquerque et al., 2022 [DOI](#)).

Pathological analysis of inflammatory cell infiltrates in tissue biopsies

Tissue biopsies were available from a subset of liquid biopsies evaluated for the gfDNA content, allowing the investigation of inflammatory cell infiltrates. In such cases, pathological tissue analyses were done independently by two expert cancer pathologists, both specialized in upper gastrointestinal tract neoplasia, who were blinded to the gfDNA concentration results from the liquid biopsies. Discrepant cases for the immune infiltration levels (n=3) were revised and discussed, and a consensus was achieved. After the microscopic identification of tumor cells and the histological classification of gastric cancer, the presence and intensity of inflammatory cell

infiltrates in the neoplastic lesions were also evaluated, as well as their patterns in the tumor stroma (e.g., diffuse, microscopic lymphoid aggregates, lymphoid aggregates with the formation of hyperplastic germinal centers). We have set the predominant cellular composition of the inflammatory infiltrates, and quantified the infiltrate/tumor stroma ratios as follows: (i) minimal/rare – rare inflammatory cells in the tumor stroma; (ii) moderate – intermediate cellularity, no distortion of the epithelial component and without formation of luminal micro-abscesses, and (iii) intense/elevated – large amounts of inflammatory cells, making it difficult to visualize the stroma, with the formation of luminal micro-abscesses and/or epithelial thinning. We have also evaluated the presence of lymphoid aggregates, necrosis/ulcer and neutrophilic infiltrates. Areas directly related to ulcer formation and/or granulation tissue were not considered in the polymorphonuclear component assessment.

Statistical analysis

Baseline gFDNA values for patient groups are given as a mean. Analyses were performed in the R statistical environment (version 4.2.0), as described (R Core Team, 2022 [\[1\]](#)). Statistical analyses were carried out by using one-way ANOVA, Tukey's HSD, Mann-Whitney test and the Kruskal-Wallis test for paired or multiple comparisons. Survival data were collected for all subjects and the cutoffs on gFDNA concentrations, determined to separate survival groups, were calculated using the log-rank test (function `maxstat.test` from R package `maxstat`; Hothorn, 2017 [\[2\]](#)). Distributions of gFDNA concentrations for the four groups of patients (i.e., normal, peptic ulcer diseases, preneoplastic diseases, and gastric cancer) were compared by using the two samples Wilcoxon test, and p-values were adjusted by the Benjamini-Hochberg procedure (i.e., False Discovery Ratio, FDR). Logistic Regression Analysis was used to determine Receiver Operating Characteristics (ROC) and Area Under the Curve (AUC). Results were considered statistically significant when $p \leq 0.05$.

Results

Patient groups and diagnoses

Gastric fluids were collected and gFDNA was extracted from a total of 1,056 subjects. After excluding 115 cases with diagnosis of hepatic portal hypertension, partial or total gastrectomy, esophagectomy, or other non-gastric cancer malignancies (e.g., esophageal squamous cell carcinoma, duodenal adenocarcinoma, Hodgkin's disease, well-differentiated neuroendocrine tumor, and metastatic breast cancer, among others), a total of 941 patients (89.1%) remained for gFDNA analysis. These patients were grouped according to the EGD findings as: normal EGD ($n = 10$, 1.1%), non-malignant peptic diseases ($n = 596$, 63.3%) including erosive gastritis, erosive and non-erosive esophagitis with or without hiatal hernia, infectious or inflammatory disorders), preneoplastic conditions ($n = 99$, 10.5%) including atrophic gastritis, gastric intestinal metaplasia, dysplasia, Barret's esophagus, or gastric adenocarcinoma ($n = 236$, 25.1%).

Diagnostic group, age, sex, and body-mass index (BMI) of the participants are shown in **Table 1** [\[3\]](#).

Gastric fluid DNA concentrations according to age, pH and body-mass index

No significant variations in gFDNA concentration were found by age, pH of gastric fluid, or BMI when examined by group amongst gastric cancer patients or non-cancer controls. Regarding age, subjects were clustered in the three groups (<45, 45-60, and >60 years old) and no significant differences were found between subjects with ($p = 0.26$) or without ($p = 0.14$) gastric cancer. Similarly, no difference was found by pH analysis of gastric fluid (grouped as $pH < 7$, $pH = 7$, or $pH > 7$); we found p-values of 0.60 and 0.90 respectively for gastric cancer and non-cancer controls. For BMI analysis, subjects were grouped as underweight (BMI < 18.5), normal (BMI between 18.5 - 24.9), overweight (BMI between 25 - 29.9), or obese (BMI ≥ 30). Again, no differences in gFDNA concentrations were found by BMI in individuals with gastric cancer ($p = 0.59$) or non-cancer controls ($p = 0.25$).

		Non-gastric cancer* (n=705, 74.9%)	Gastric cancer (n=236, 25.1%)
Age	≤45	130 (18.4%)	34 (14.4%)
	45-60	224 (31.8%)	83 (35.2%)
	≥60	343 (48.7%)	119 (50.4%)
	Missing data	8 (1.1%)	0
Sex	Male	330 (46.8%)	150 (63.6%)
	Female	367 (52.1%)	86 (36.4%)
	Missing data	8 (1.1%)	0
**BMI	Underweight	10 (1.4%)	9 (3.8%)
	Normal weight	206 (29.2%)	91 (38.6%)
	Overweight	310 (44%)	84 (35.6%)
	Obese	177 (25.1%)	51 (21.6%)
	Missing data	2 (0.3%)	1 (0.4%)

* Includes all patients diagnosed as presenting normal mucosa, with peptic diseases or pre-neoplastic lesions.

**Body-mass index (BMI): Underweight, BMI<18.5; Normal weight, BMI=18.5-24.9; Overweight, BMI=25-29.9; Obese, BMI≥30

Table 1. Clinical and demographic attributes of participating subjects (n=941)

gfDNA concentrations and sex

Next, we evaluated whether gfDNA concentrations vary according to sex in patients with gastric cancer and subjects without cancer. For the non-cancer cohort, we observed elevated gfDNA concentrations in females versus males ($p = 7.44e^{-4}$), but no differences were seen between males and females in the cancer group ($p = 0.68$) (**Figure 1**). Therefore, when we compared gfDNA concentration between cancer versus non-cancer subjects, considering sex, we found striking differences for men and for women, respectively, $p = 8.75e^{-12}$ and $p = 1.20e^{-12}$ (**Figure 1**).

As some studies have suggested that the use of proton pump inhibitors (PPI) is more frequent in women (Linseisen et al., 2022), we evaluated possible links between gfDNA concentrations and the reported history of current, previous, or no PPI use. In general, we saw no impact of PP on gfDNA concentrations. However, for the group of non-cancer subjects that used PPI ($n=410$), increased gfDNA concentrations were seen for women ($n=206$), as compared to men ($n=204$) ($p = 1.51e^{-4}$), apparently reflecting the sex-related results shown in **Figure 1**, between males and females, unrelated to PPI-use. Moreover, gfDNA concentrations were not different for subjects that reported current and previous administration, as well as no PPI administration, for females with or without cancer ($n=55$, $n=6$ and $n=25$; $n=111$, $n=45$ and $n=206$, respectively), as well as for males with or without cancer diagnosis ($n=60$, $n=19$ and $n=62$; $n=75$, $n=51$ and $n=209$, respectively, with corresponding p-values of $p = 0.92$; $p = 0.07$; $p = 0.24$ and $p = 0.71$).

Analysis of bacteria to human ratios in gfDNA

As the gfDNA origin is likely to be a sum of human DNA (including epithelial, stromal, and immune cells, tumor and non-tumor cells), combined with microbiota-derived DNA (transient, saliva-derived microorganisms, the resident microbiota of the stomach and other upper digestive tract structures), we performed a quantitative analysis of bacteria and human DNA for a subset of samples ($n=180$), based on human-to-bacteria DNA ratios, as recently described (Albuquerque et al., 2022). The samples selected for the analysis of this subset analysis included all 10 controls without any pathologies detected, as well as samples from the peptic diseases group ($n=51$), preneoplastic conditions ($n=55$) and the gastric cancer group ($n=64$). This analysis revealed no significant differences in the ratios of human- or bacteria-derived DNA found in the gfDNA (**Suppl. Table 1**).

gfDNA concentrations are increased in gastric cancer patients compared to non-cancer controls

We first determined the concentration of gfDNA from subjects with different diagnoses after EGD and found the following values for the groups: normal mucosa and peptic diseases – mean 10.77 ng/ μ L; 95% CI: 9.23 to 12.33; $n=606$; preneoplastic conditions - mean 10.10 ng/ μ L; 95% CI: 7.59 to 12.60; $n=99$ and gastric cancer - mean 26.86 ng/ μ L; 95% CI: 20.05 to 33.79; $n=236$. Notably, mean concentration of gfDNA was significantly different among the groups ($p=3.61e^{-12}$). gfDNA concentration was higher in gastric cancer as compared to the combined group of normal EGD and peptic disease patients ($p=9.55e^{-13}$), as well as to the preneoplastic group ($p=1.10e^{-5}$). In contrast, gfDNA concentrations were not different between the non-cancer groups ($p=0.89$) (**Figure 2A**).

For gastric cancer patients, we evaluated possible correlations between gfDNA concentrations and clinicopathological variables such as tumor-stage and histopathological grades. In general, increments in gfDNA were found in subjects as the disease progresses (**Figure 2B**). Notably, the gfDNA mean concentration for early-stage disease (tumor stages - T0 + Tis + T1 + T2, or T0-T2 $n=65$) was 15.12 ng/ μ L; 95% CI: 9.73 to 20.50, increasing in T3 (25.66 ng/ μ L; 95% CI: 19.46 to 31.85; $n=91$; $p = 5.97e^{-4}$) and T4 stages (38.12 ng/ μ L; 95% CI: 15.02 to 61.22; $n=63$; $p = 0.06$).

Other comparisons were not significant, including gfDNA concentrations for patients with localized ($n=165$) versus metastatic disease ($n=71$). gfDNA concentrations could not differentiate tumor histopathological grades ($p>0.05$), or Lauren gastric cancer subtypes: diffuse ($n=97$), intestinal ($n=93$), or mixed ($n=25$) ($p = 0.28$) (data not shown).

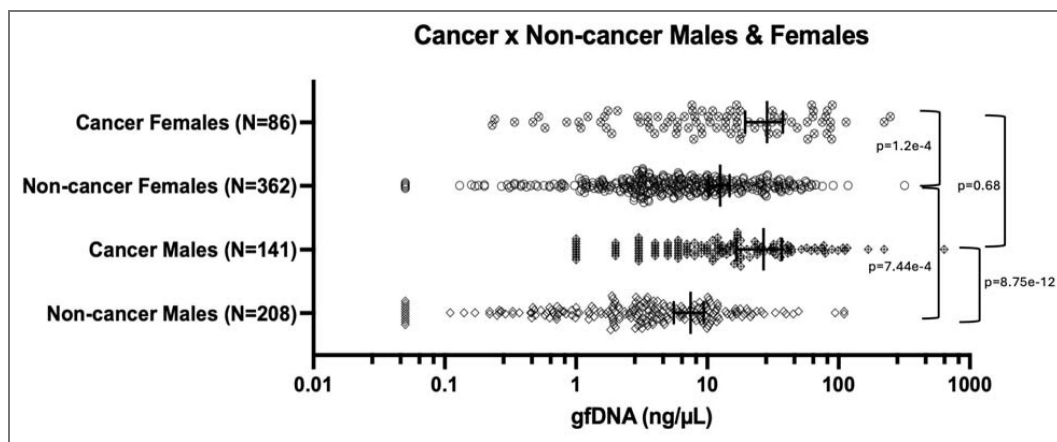


Figure 1. gfDNA concentrations observed in all 941 subjects, grouped according to sex, and presence of gastric cancer.

Non-cancer subjects have normal gastric mucosa or minor peptic diseases (excludes pre-neoplastic disease). Bars indicate mean gfDNA concentration (ng/μL) and 95% confidence intervals.

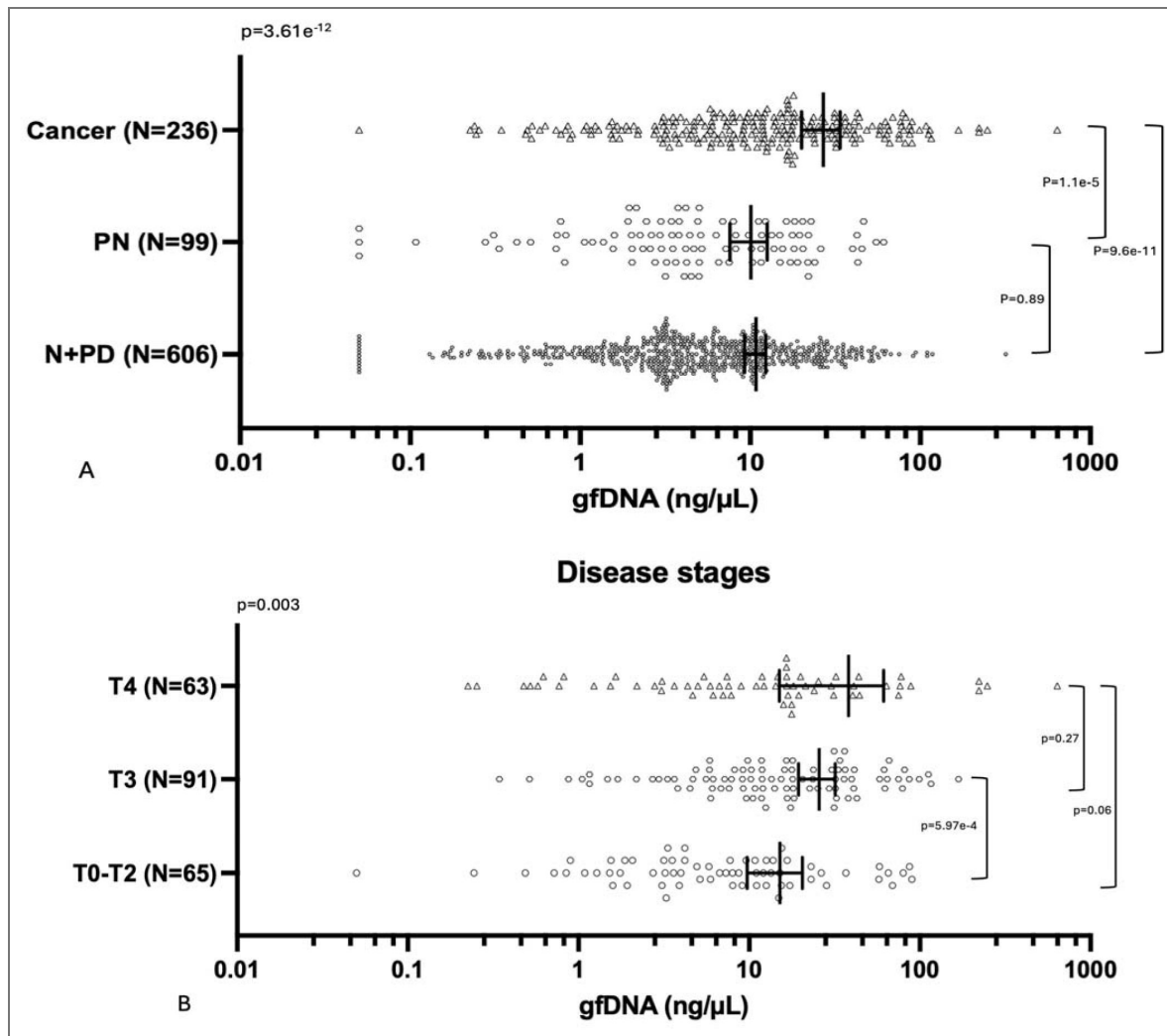


Figure 2. gfDNA concentration according to disease diagnosis and tumor stage.

A) gfDNA concentration (ng/μL) in patients with no endoscopic findings: Normal (N) or presenting minor peptic diseases (PD) – N+PD; preneoplastic conditions (PN), or gastric cancer (Cancer). B) gfDNA concentrations (ng/μL) for early-stage disease patients – T0-T2 (T0 + T1 + T2), as compared to more advanced disease stages T3 and T4. Bars indicate mean gfDNA concentration (ng/μL) and 95% confidence intervals.

gfDNA correlates with the presence of inflammatory cells infiltrates and gastric cancer diagnosis and survival

Our liquid biopsy results revealed increased gfDNA concentrations in gastric cancer patients as compared to non-cancer controls. To explore whether gfDNA concentrations could be used to support gastric cancer diagnosis, we analyzed the ROC curve of gfDNA between gastric cancer versus non-cancer controls. We observed a statistically significant curve ($AUC=0.66$; $p=5.23e^{-10}$), with a fair capability of detecting true positive gastric cancer patients by simply considering gfDNA concentrations (**Suppl. Figure 1A** [↗](#); **Suppl. Table 1** [↗](#)), indicating its fair diagnostic support value. We also found a statistically significant curve of gfDNA between cancer/preneoplastic conditions and peptic diseases/normal ($AUC=0.61$; $p=6.86e^{-7}$) (**Suppl. Figure 1B** [↗](#)).

We next evaluated if gfDNA concentrations would correlate with disease prognosis. Well-annotated clinical records were collected for all subjects to determine gfDNA concentrations cutoff values that would split the patients into groups with contrasting overall survival through a log-rank test. By using this approach, a gfDNA cutoff of 1.28 ng/ μ L was set and gastric cancer patients presenting gfDNA concentrations above this threshold had better overall survival ($n=163$; $p=0.019$), even more significant after patients with metastatic tumors were not considered ($n=148$; $p=0.014$) (**Figure 3A** [↗](#)). Finally, to further investigate the possible link between patient survival and gfDNA concentration, we examined a subset of gastric cancer patients ($n=32$), for which pathology slides were available, to investigate possible correlations between gfDNA concentrations and the intensity of inflammatory cell infiltrates (**Figure 3B** [↗](#)). For this we managed to include 8 samples with gfDNA below the threshold, and 24 above it. Even with a small sample set, this result was also significant ($p=0.001$) (**Figure 3C** [↗](#) and **Suppl. Table 3** [↗](#)), suggesting a possible mechanism to support the correlation of gfDNA concentration and gastric cancer survival data (**Figure 3A** [↗](#)).

Discussion

Gastric cancer remains a leading cause of tumor mortality worldwide, a clinical feature attributable at least in part to its often-late stage at diagnosis and/or lack of prognostic factors to enable personalized disease management. However, with varying access to subspecialists a role for gfDNA may help in the clinical management of patients. Early gastric cancer detection may often be delayed during EGD if biopsies samples are either taken too superficially or if the tissue is too scanty for meaningful pathological analysis. Thus, the translational application of a biomarker that provides support to gastric cancer diagnosis and prognosis would be useful, especially in a less specialized outpatient ambulatory or inpatient hospital settings.

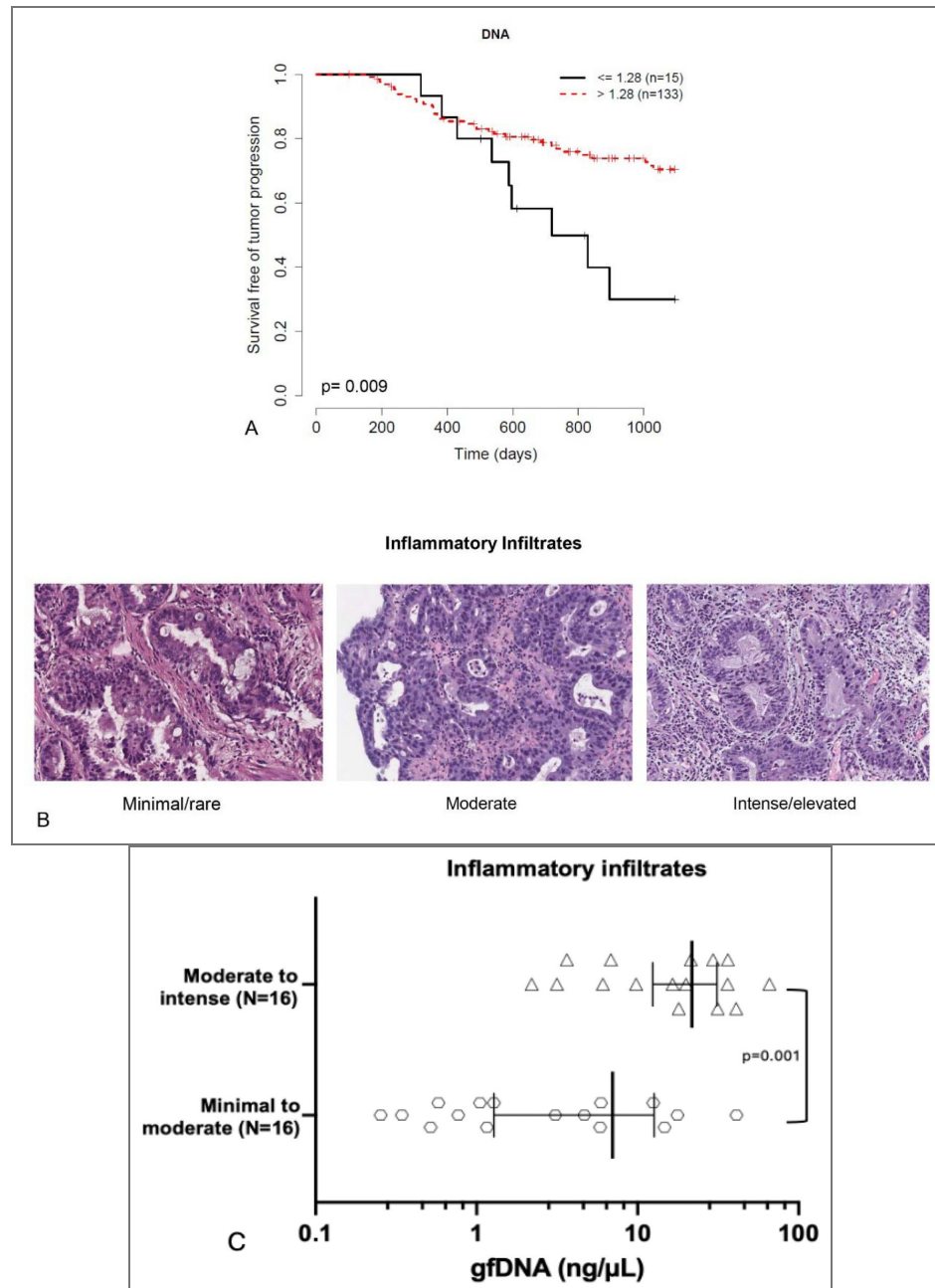
In 1977, Leon et al. published a seminal paper showing that cancer patients had higher levels of cell-free DNA (cfDNA) in their blood compared to healthy individuals, suggesting its use as a potential cancer biomarker. Since then, numerous studies confirmed these findings and have explored the potential of using cfDNA for cancer diagnosis, prognosis, and monitoring treatment response, including gastric cancer. Qian et al.(2017) [↗](#) analyzed cfDNA in serum samples of subjects with gastric dysfunctions, including gastric cancer, and found that cancer patients presented elevated cfDNA when compared to benign gastric disease or controls.

However, the diluted amounts of tumor-derived/tumor-microenvironment DNA remains minute in the peripheral circulation. This prompted us to determine DNA concentrations in the gastric fluids of subjects undergoing EGD to investigate non-specific digestive symptoms.

Patients included those with diverse peptic diseases (controls), pre-neoplastic lesions as well patients with confirmed gastric cancer. We reasoned that liquid biopsies of gastric fluid incidentally obtained during EGD would offer diagnostic support and would possibly offer clinically useful information. Whereas EGD is an invasive procedure, most of these patients will receive at least one diagnostic EGD during their initial workup and tumor staging, and gastric fluids collected at this moment can be useful. However, prior studies of gastric fluids have

Figure 3. gfDNA analysis according to patient survival and infiltration of inflammatory cells.

(A) For a total of 148 non-metastatic GC patients, cutoff gfDNA concentrations determined by a log-rank test (1.28 ng/ μ L) discriminate survival free of tumor progression ($p = 0.009$). (B) For a subset of 32 cases, for which representative biopsy H&E slides were available, inflammatory cell infiltrates in the gastric tumors were inspected by experienced pathologists and (C) have shown that more intense inflammatory infiltrates in tumors are characteristic of subjects with higher gfDNA concentrations ($p = 0.001$). Bars indicate mean gfDNA concentration (ng/ μ L) and 95% confidence intervals.



heretofore been limited to the evaluation of certain molecules, such as proteins, RNAs or for the analysis of mutations in specific genes (Wu, Chung, 2013; Shao et al., 2014; Pizzi et al., 2019 [Pizzi et al., 2019](#)). Here we show that the simple evaluation of gfdNA concentration serves as a convenient and promising biomarker to support gastric cancer diagnosis and staging and correlates with immune cell infiltrates. In a large cohort, we show that increased DNA concentrations in gastric fluids may differentiate gastric cancer patients from individuals without cancer, independent of age, BMI, sex, PPI use, or Lauren's histological subtypes. Increasing amounts of gfdNA were significantly associated with gastric cancer diagnosis, as well as advanced tumor stages.

Of note, there were several interesting observations from our results. Overall, we found that T4 tumors had gfdNA concentration similar to T3 tumors (T3: 25.66 ng/μL; 95% CI: 19.46 to 31.85; and T4: 38.12 ng/μL; 95% CI: 15.02 to 61.22; $p=0.27$) and the difference between T4 and earlier stages (T2 and below) was non-significant ($p = 0.065$). One might speculate that this may be due to the higher variability of gfdNA concentration seen for T4 (**Figure 2B** [Figure 2B](#)) that could be intrinsic to the T4 stage, as some tumors are more invasive than others, and if the tumor grows through the gastric wall and the serosa (Wu, Chung, 2013) it might perhaps be shedding DNA into adjacent peritoneal structures instead of just the gastric cavity, leading to a reduction of tumor-derived DNA in the gastric cavity. This contrasts with T0-T3 tumors that appear to continuously and preferentially release DNA inside the gastric cavity. In agreement with this possibility, we found that advanced tumors have increased gfdNA concentration as compared to early stages, a finding that might suggest that the origin of the gfdNA could perhaps derive from the tumor, but also from the tumor environment, from immune cell infiltrates, and the microbiota. However, our quantitative analysis of bacterial and human DNA in the gfdNA for a set of samples ($n=180$), including gastric cancer samples ($n=64$), showed no clear trends towards an increased bacterial DNA content, rendering that possibility unlikely (**Suppl. Table 2** [Suppl. Table 2](#)). We also observed an intriguing correlation between gfdNA concentration and survival, with gastric cancer patients presenting lower gfdNA concentrations (set at ≤ 1.28 ng/μL) associated with a worse prognosis. We envisaged that this observation could be associated with poorer immune responses (immunologically “cold tumors”), which was supported by the finding that patients with higher gfdNA concentrations (set at >1.28 ng/μL) more often presented with moderate-to-intense inflammatory infiltrates, whereas individuals with lower gfdNA usually presented minimal-to-moderate inflammatory infiltrates ($p=0.001$). Notably, two-thirds (16 out of 24) of the cases with increased gfdNA had moderate/intense/elevated immune cells infiltrates and all cases within this classification of inflammatory infiltrates were in the group of increased gfdNA (**Suppl. Table 3** [Suppl. Table 3](#)). This suggests that the elevated gfdNA concentration observed in subjects with better outcomes may originate from two primary sources: an active population of infiltrating immune cells, together with the tumor cells eliminated by these immune infiltrates.

The results presented here suggest wide implications for the translational application of liquid biopsies with gfdNA concentration analysis in the diagnosis, staging and management gastric cancer patients. The most obvious barrier to the translational application of liquid biopsies and gfdNA measurement in the routine clinical setting is the need of an EGD to access it. However, the use of gfdNA may hold the highest benefit by increasing the value of information obtained from the initial EGD, which is --as mentioned previously-- a procedure that all patients with gastric cancer will eventually undergo multiple times during their disease workup. Indeed, the suction of gastric fluids during endoscopy is readily performed during EGD, when the gastric walls must be free of fluids for a better visualization of the mucosa. Thus, the convenient incorporation of this liquid biopsy approach during EGD is cost-effective and a minor additional step to the procedure. The benefit is that the continuous proximity of gastric fluids to the tumor site portends a much-reduced dilution of gfdNA, some of it being tumor-derived DNA, which allows a better representation of tumor-related DNA alterations. In a comparative sense, liquid biopsy of gastric fluid with gfdNA measurement is similar to obtaining DNA from peritoneal lavage of patients with endometrial cancer, but indeed much harder to obtain, in which peritoneal fluid has higher mutant allelic fractions as compared to plasma (Mayo-de-las-Casas, 2020 [Mayo-de-las-Casas, 2020](#)), similar to previous finding in gastric cancer in our early work (Pizzi et al., 2019 [Pizzi et al., 2019](#)).

Much interest has been given to developing new tools for detecting diagnostic, prognostic, and predictive markers in gastric cancer (Mayo-de-las-Casas, 2020). The area under the receiver operating characteristic (AUROC) analysis for liquid biopsy of gastric fluid with gfdNA analysis yielded a discrimination capability of 0.66, which is comparable to serological tumor biomarkers such as CEA (0.68), CA72-4 (0.67), or CA19-9 (0.64) that are currently used in standard oncological practice (Tahara, Arisawa, 2015; Yu et al., 2016; Lin et al., 2020). Moreover, the monitoring of gfdNA concentration may also be valuable for a closer follow up and better diagnostic accuracy of patients with potentially premalignant diseases and/or at higher genetic/familial risk of developing gastric cancer. One might also speculate that our findings could perhaps also apply to other upper gastrointestinal malignant tumors such as esophageal cancer and/or gastroesophageal junction neoplasia, and future studies should rigorously evaluate those open possibilities.

This study represents an initial evaluation of the prognostic impact of gfdNA concentration in human subjects diagnosed with gastric cancer, as compared to other conditions such as pre-malignant lesions and peptic diseases. The participants were referred to diagnostic EGD and their distribution in the diagnostic groups reflect the real-life case numbers seen in a single institution, a dedicated cancer center. Before receiving sedation for the EGD, participants were invited to join the study. Therefore, samples were collected prospectively, before diagnosis, from all subjects and no attempts to balance the groups were made to avoid introducing any potential selection bias. Limitations also include the primer sets that allow a partial quantification of some bacterial groups (i.e., microbiome analysis), the limited number of pathology slides available from gastric cancer patients to investigate immune cell infiltrates (n=32), and the lack of a more comprehensive approach that would also allow a cell type-specific deconvolution and a better explanation of the functional mechanisms that lead to increased gfdNA concentrations. The diagnostic potential of elevated gfdNA levels was not explored together with other known biomarkers such as CEA or CA19-9. Our data suggest that gfdNA is not only derived from tumor cells but may have expressive amounts of DNA derived from infiltrating immune cells. The precise determination of the presumed cell-of-origin of this increased gfdNA remains to be unequivocally demonstrated as it would have to rely on methods not used in this study, such as RNA-Seq and DNA-seq.

Moreover, the large patient cohort analyzed in this study was derived from a single specialized institution (a tertiary cancer center), and it is therefore unclear whether our findings might be broadly generalized; this open question should be addressed in further confirmatory studies from other hospital types and geographical regions.

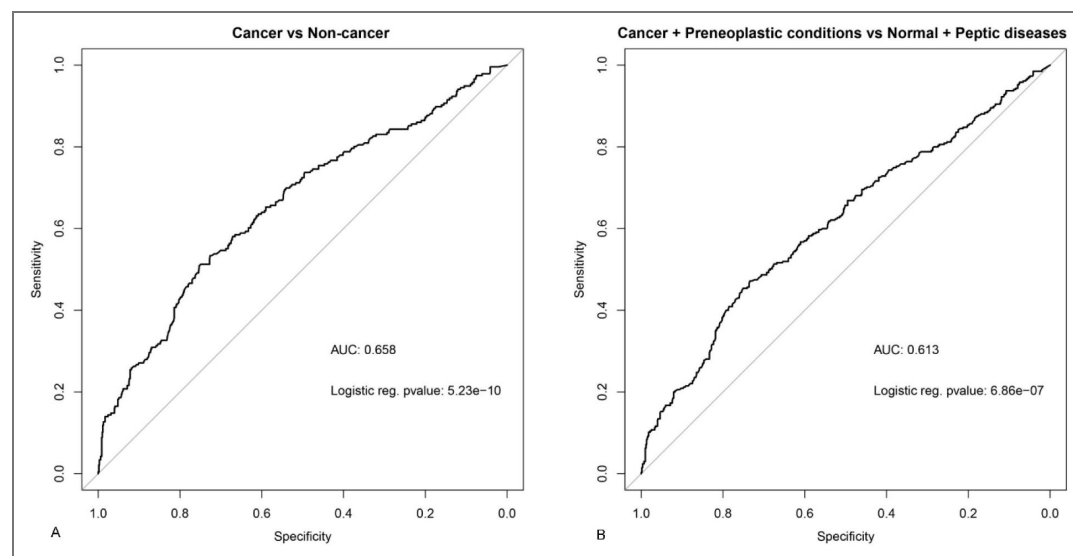
Finally, for considering subjects with an established gastric cancer diagnosis, gfdNA has the potential to serve as a rapid, low-cost surrogate marker of the tumor immune microenvironment, whose characterization may have implications for predicting chemotherapy and immunotherapy response (Jiang et al., 2018; Jiang et al., 2019; Chen et al., 2022; Espinosa-Carrasco et al., 2024), and prognostic repercussions that, in the future, could be useful to treatment planning and risk-adapt therapeutic strategies. As we have gathered no data on gfdNA changes over time, conclusions on its utility for monitoring treatment response and/or predicting recurrence are currently limited, but warrant further investigation.

Given that gastric cancer has currently no suitable tumor prognostic biomarker, one hopes that the initial results of this report will encourage the liquid biopsy of EGD-collected gastric fluids to evaluate gfdNA concentrations towards clinic-ready translational applications. Further gastric fluid studies may help revolutionize gastric cancer care especially in places where specialties may not be readily available. Also, it may assist as a first line exam that can be done to help detect cancer cases, with prognostic value. Whereas more research is needed, early data indicates this may be a way to provide access and availability for primary and recurrent gastric cancer.

Data availability

All data supporting the findings of this study are contained within the manuscript. The corresponding author will gladly consider sharing additional materials or datasets upon request to support further scientific inquiry.

Supplementary information



Supplementary Figure 1. Analysis of Receiver Operating Characteristics (ROC) and Area Under the Curve (AUC) for gfdNA levels between cancer patients and non-cancer individuals. A) ROC curve of gfdNA between the cancer patients and non-cancer individuals. B) ROC curve of gfdNA between cancer versus non-cancer and cancer + preneoplastic conditions versus peptic diseases + normal.

Supplementary Table 1. Performance metrics of the ROC/AUC curves shown in Suppl. Figure 1 [↗](#), panels A and B.

Cancer vs. non-cancer thresholds (Suppl. Fig. 1A)				Cancer + pre-neoplastic lesions vs. normal + peptic diseases (Suppl. Fig. 1B)			
Threshold	Specificity	Sensitivity	Youden_J	Threshold	Specificity	Sensitivity	Youden_J
11.81	0.7532	0.5085	0.2617	10.75	0.7343	0.4716	0.2060
11.75	0.7489	0.5127	0.2616	11.55	0.7508	0.4537	0.2046
11.65	0.7475	0.5127	0.2602	10.65	0.7310	0.4716	0.2027
10.75	0.7262	0.5339	0.2601	11.65	0.7541	0.4478	0.2019
11.86	0.7532	0.5042	0.2574	10.9	0.7360	0.4657	0.2016
10.65	0.7234	0.5339	0.2573	11.485	0.7475	0.4537	0.2013
10.55	0.7220	0.5339	0.2559	10.55	0.7294	0.4716	0.2010
11.55	0.7418	0.5127	0.2546	11.81	0.7591	0.4418	0.2009
10.455	0.7163	0.5381	0.2544	11.465	0.7459	0.4537	0.1996
10.9	0.7277	0.5254	0.2531	11.75	0.7541	0.4448	0.1989

Supplementary Table 2. Analysis of CT values for bacteria and human target genes.

	n	Median CT	p
CT 16S-V1			
Normal	10	17.27	0.21
Peptic diseases	51	18.60	
Preneoplastic conditions	55	17.72	
Cancer	64	18.92	
CT β -ACT			
Normal	10	25.29	0.07
Peptic diseases	51	26.21	
Preneoplastic conditions	55	27.01	
Cancer	64	27.46	
Ratio 16S-V1/ β -ACT			
Normal	10	0.68	0.83
Peptic diseases	51	0.71	
Preneoplastic conditions	55	0.66	
Cancer	64	0.69	

Supplementary Table 3. gfdNA concentration and intensity of inflammatory cell infiltrates.

Inflammatory infiltrates	Lower gfdNA (<1.28ng/ μ L) N=8 patients	Higher gfdNA (>1.28ng/ μ L) N=24 patients
Minimal/rare	0	8.3% (n=2)
Minimal/rare to moderate	62.5% (n=5)	12.5% (n=3)
Moderate	37.5% (n=3)	12.5% (n=3)
Moderate to intense/elevated	0	50% (n=12)
Intense/elevated	0	16.6% (n=4)

Acknowledgements

Authors acknowledge the support given by the institutional Tumor Bank from A. C. Camargo Cancer Center. We thank Dr. Christian Abnet and Dr. M. Constanza Camargo (NCI, NIH, USA) for their critical review of this manuscript.

Additional information

Author Contributions

Collected samples and clinical data: FCC, AGP, CZS, LBCS, GPB, FAP, LLSA

Concepted and performed analysis: FCC, TFB, AD, RD, ITS, DNN, EDN

Pathology: APB, GOS, WAN

Data analysis and statistics: FCC, TFB, AD, RDD, ITS, RP, WA, DNN, EDN

Discussion of findings: FCC, TFB, AGP, CZS, LGVC, ML, HI, SL, HSW, FJFV, ITS, RC, RP, WA, DNN, EDN

Writing & editing: FCC, TFB, AGP, CZS, LGVC, ML, HI, SL, HSW, FJFV, ITS, RC, RP, WA, DNN, EDN

Final manuscript review and approval: All authors

Project supervision: EDN

Funding Statement

This work was supported in part by grants from the Sao Paulo State Research Foundation (FAPESP; 14/2687-0, 18/14267-2, 18/02972-3; PI: ED-N) and from the Science and Technology Department of the Health Ministry (DECIT, PRONON, SIPAR 2500.035-167/2015-23; PI: ED-N) of the Ministry of Health, both in Brazil. RP and WA received research awards and support from the Levy-Longenbaugh Donor-Advised Fund.

Funding

Funder	Grant reference number	Author
Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP)	2014/2687-0	Emmanuel Dias-Neto
Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP)	2018/14267-2	Emmanuel Dias-Neto
Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP)	2018/02972-3	Emmanuel Dias-Neto
Ministério da Saúde (Ministry of Health)	PRONON/DECIT, SIPAR 2500.035-167/2015-23	Emmanuel Dias-Neto

Author ORCID iDs

Francine Carla Cadoná: <http://orcid.org/0000-0002-6761-9743>

Thais F Bartelli: <http://orcid.org/0000-0002-1373-0112>

Adriane Graicer Pelosof: <http://orcid.org/0000-0001-9813-5832>

Claudia Zitron Sztokfisz: <http://orcid.org/0000-0001-9364-8460>

Adriana Passos Bueno: <http://orcid.org/0000-0002-8671-2847>

Luana Batista do Carmo dos Santos: <http://orcid.org/0000-0001-5104-3914>

Gabriela Pereira Branco: <http://orcid.org/0000-0002-3120-4116>

Gabriel Oliveira dos Santos: <http://orcid.org/0000-0002-1891-9221>

Warley Abreu Nunes: <http://orcid.org/0000-0003-0927-5636>

Fernanda Araújo Pintor: <http://orcid.org/0000-0002-8643-1770>
Laís Lie Senda de Abrantes: <http://orcid.org/0000-0003-2178-8805>
Alexandre Defelicibus: <http://orcid.org/0000-0002-3484-6317>
Luiz Gonzaga Vaz Coelho: <http://orcid.org/0000-0002-8721-7696>
Marcis Leja: <http://orcid.org/0000-0002-0319-8855>
Haejin In: <http://orcid.org/0000-0003-3267-6585>
Sharon Li: <http://orcid.org/0000-0002-1372-6631>
Howard Hochster: <http://orcid.org/0000-0002-9893-1529>
Felipe José Fernandez Coimbra: <http://orcid.org/0000-0001-5068-0639>
Rodrigo D Drummond: <http://orcid.org/0000-0002-9000-8115>
Israel Tojal da Silva: <http://orcid.org/0000-0002-4687-1499>
Ravi J Chokshi: <http://orcid.org/0000-0003-2547-4942>
Renata Pasqualini: <http://orcid.org/0000-0002-4155-9324>
Wadih Arap: <http://orcid.org/0000-0002-8686-4584>
Diana Noronha Nunes: <http://orcid.org/0000-0003-0751-4701>
Emmanuel Dias-Neto: <http://orcid.org/0000-0001-5670-8559>

Author notes

Thais F Bartelli: Present address: Department of Clinical Cancer Prevention, The University of Texas M. D. Anderson Cancer Center, Houston, USA.

Competing interests: RP and WA are founders, equity stockholders, and scientific advisors of PhageNova Bio and of MBrace Therapeutics. PhageNova Bio and MBrace Therapeutics partially support their academic research laboratory through Sponsored Research Agreements. These relationships are currently managed according to established institutional conflict of interest policies of Rutgers, The State University of New Jersey. The work presented here is unrelated and falls outside of the scope from these commercial entities. Authors declare no potential conflicts of interest.

References

- Albuquerque G. E., et al.** (2022) Evaluation of Bacteria and Fungi DNA Abundance in Human Tissues. *Genes (Basel)* **13**:237 <https://doi.org/10.3390/genes13020237>
- Bartelli T. F., et al.** (2019) Genomics and epidemiology for gastric adenocarcinomas (GE4GAC): a Brazilian initiative to study gastric cancer. *Appl Cancer Res* **39**:1-4 <https://doi.org/10.1186/s41241-019-0085-1>
- Chen Y., et al.** (2022) Predicting response to immunotherapy in gastric cancer via multi-dimensional analyses of the tumour immune microenvironment. *Nat Commun* **13**:4851 <https://doi.org/10.1038/s41467-022-32540-1>
- Espinosa-Carrasco G., et al.** (2024) Intratumoral immune triads are required for immunotherapy-mediated elimination of solid tumors. *Cancer Cell* **19**:S1535-6108(24)00193-4 <https://doi.org/10.1016/j.ccell.2024.01.010>
- Forsheew T., et al.** (2012) Noninvasive identification and monitoring of cancer mutations by targeted deep sequencing of plasma. *DNA Sci Transl Med* **4**:136ra68 <https://doi.org/10.1126/scitranslmed.3003726>
- Fukuda N., et al.** (2011) Prognostic factors of T4 gastric cancer patients undergoing potentially curative resection. *World J Gastroenterol* **17**:1180-1184 <https://doi.org/10.3748/wjg.v17.i9.1180>
- Globocan/IARC** (2022) Globocan 2020 – World fact sheet. <https://gco.iarc.who.int/media/globocan/factsheets/cancers/7-stomach-fact-sheet.pdf>
- Hamashima C., et al.** (2013) A community-based, case-control study evaluating mortality reduction from gastric cancer by endoscopic screening in Japan. *PLoS One* **396**:635-648 <https://doi.org/10.1371/journal.pone.00635648>

- Hothorn T (2017) maxstat: Maximally Selected Rank Statistics R package version 0.7-25. <https://CRAN.R-project.org/package=maxstat>
- Jiang Y., et al. (2019) Tumor Immune Microenvironment and Chemosensitivity Signature for Predicting Response to Chemotherapy in Gastric Cancer. *Cancer Immunol Res* **7**:2065-2073 <https://doi.org/10.1158/2326-6066.CIR-19-0260>
- Jiang Y., et al. (2018) ImmunoScore Signature: A Prognostic and Predictive Tool in Gastric Cancer. *Ann Surg* **267**:504-513 <https://doi.org/10.1097/SLA.0000000000002491>
- Kim G. H., Jung H. Y (2021) Endoscopic Resection of Gastric Cancer. *Gastrointest Endosc Clin N Am* **31**:563-579 <https://doi.org/10.1016/j.giec.2021.04.005>
- Leon S. A., et al. (1977) Free DNA in the serum of cancer patients and the effect of therapy. *Cancer Res* **37**:646-50 <https://doi.org/10.1158/0008-5472.CAN-77-037>
- Lin R., et al. (2020) Genome-wide DNA methylation profiling identifies epigenetic signatures of gastric cardiac intestinal metaplasia. *J Transl Med* **18**:292 <https://doi.org/10.1186/s12967-020-02457-9>
- Matsumoto S., Yoshida Y. (2014) Efficacy of endoscopic screening in an isolated island: A case-control study Indian. *J Gastroenterol* **33**:46-49 <https://doi.org/10.1007/s12664-014-0416-6>
- Mayo-de-las-Casas C., et al. (2020) Detection of somatic mutations in peritoneal lavages and plasma of endometrial cancer patients: A proof-of-concept study. *Int J Cancer* **147**:277-284 <https://doi.org/10.1002/ijc.32777>
- Pizzi M. P., et al. (2019) Identification of DNA mutations in gastric washes from gastric adenocarcinoma patients: Possible implications for liquid biopsies and patient follow-up. *Int J Cancer* **145**:1090-1097 <https://doi.org/10.1002/ijc.32114>
- Qian C., et al. (2017) Alu-based cell-free DNA: a novel biomarker for screening of gastric cancer. *Oncotarget* **8**:54037-54045 <https://doi.org/10.18632/oncotarget.17671>
- R Core Team (2022) *R: A language and environment for statistical computing* Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org>
- Rückert-Eheberg I. M., et al. (2022) Who gets prescriptions for proton pump inhibitors and why? A drug-utilization study with claims data in Bavaria, Germany, 2010–2018. *Eur J Clin Pharmacol* **78**:657-667 <https://doi.org/10.1007/s00228-022-03233-2>
- Tahara T., Arisawa T (2015) DNA methylation as a molecular biomarker in gastric cancer. *Epigenomics* **7**:475-86 <https://doi.org/10.2217/EPI.15.4>

Peer reviews

Reviewer #1 (Public review):

The study analyzes the gastric fluid DNA content identified as a potential biomarker for human gastric cancer. However, the study lacks overall logicity, and several key issues require improvement and clarification. In the opinion of this reviewer, some major revisions are needed:

- (1) This manuscript lacks a comparison of gastric cancer patients' stages with PN and N+PD patients, especially T0-T2 patients.
- (2) The comparison between gastric cancer stages seems only to reveal the difference between T3 patients and early-stage gastric cancer patients, which raises doubts about the authenticity of the previous differences between gastric cancer patients and normal patients, whether it is only due to the higher number of T3 patients.
- (3) The prognosis evaluation is too simplistic, only considering staging factors, without taking into account other factors such as tumor pathology and the time from onset to tumor

detection.

(4) The comparison between gfDNA and conventional pathological examination methods should be mentioned, reflecting advantages such as accuracy and patient comfort.

(5) There are many questions in the figures and tables. Please match the Title, Figure legends, Footnote, Alphabetic order, etc.

(6) The overall logicity of the manuscript is not rigorous enough, with few discussion factors, and cannot represent the conclusions drawn.

Comments on revisions:

The authors have addressed all concerns in the revision.

<https://doi.org/10.7554/eLife.107103.2.sa2>

Reviewer #2 (Public review):

Summary

The authors aimed to evaluate whether total DNA concentration in gastric fluid (gfDNA) collected during routine endoscopy could serve as a diagnostic and prognostic biomarker for gastric cancer. Using a large cohort (n=941), they reported elevated gfDNA in gastric cancer patients, an unexpected association with improved survival, and a positive correlation with immune cell infiltration.

Strengths

The study benefits from a substantial sample size, clear patient stratification, and control of key clinical confounders. The method is simple and clinically feasible, with preliminary evidence linking gfDNA to immune infiltration.

Weaknesses

(1) While the study identifies gfDNA as a potential prognostic tool, the evidence remains preliminary. Unexplained survival associations and methodological gaps weaken support for the conclusions.

(2) The paradoxical association between high gfDNA and better survival lacks mechanistic validation. The authors acknowledge but do not experimentally distinguish tumor vs. immune-derived DNA, leaving the biological basis speculative.

(3) Pre-analytical variables were noted but not systematically analyzed for their impact on gfDNA stability.

Comments on revisions:

To enhance the completeness and credibility of this research, it is essential to clarify the biological origin of gastric fluid DNA and validate these preliminary findings through a prospective, longitudinal study design.

<https://doi.org/10.7554/eLife.107103.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

"The study analyzes the gastric fluid DNA content identified as a potential biomarker for human gastric cancer. However, the study lacks overall logicity, and several key issues require improvement and clarification. In the opinion of this reviewer, some major revisions are needed:"

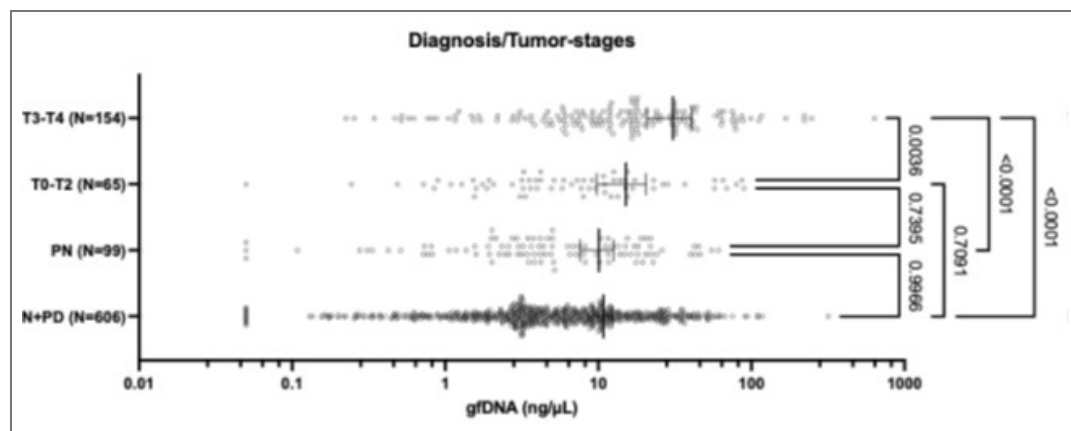
(1) "This manuscript lacks a comparison of gastric cancer patients' stages with PN and N+PD patients, especially T0-T2 patients."

We are grateful for this astute remark. A comparison of gfDNA concentration among the diagnostic groups indicates a trend of increasing values as the diagnosis progresses toward malignancy. The observed values for the diagnostic groups are as follows:

Diagnosis group	Mean gfDNA and 95% Confidence Interval
Normal mucosa and peptic diseases (N=606)	10.77 ng/μL; 95% CI: 9.23 to 12.33;
Preneoplastic conditions (N=99)	10.10 ng/μL; 95% CI: 7.59 to 12.60
T2 and below (N=65)	15.12 ng/μL; 95% CI: 9.73 to 20.50
T3 and above (N=154)	30.75 ng/μL; 95% CI: 14.24 to 36.62

Author response table 1.

The chart below presents the statistical analyses of the same diagnostic/tumor-stage groups (One-Way ANOVA followed by Tukey's multiple comparison tests). It shows that gastric fluid gfDNA concentrations gradually increase with malignant progression. We observed that the initial tumor stages (T0 to T2) exhibit intermediate gfDNA levels, which in this group is significantly lower than in advanced disease ($p = 0.0036$), but not statistically different from non-neoplastic disease ($p = 0.74$).



Author response image 1.

(2) "The comparison between gastric cancer stages seems only to reveal the difference between T3 patients and early-stage gastric cancer patients, which raises doubts about the authenticity of the previous differences between gastric cancer patients and normal patients, whether it is only due to the higher number of T3 patients."

We appreciate the attention to detail regarding the numbers analyzed in the manuscript. Importantly, the results are meaningful because the number of subjects in each group is comparable (T0-T2, $N = 65$; T3, $N = 91$; T4, $N = 63$). The mean gastric fluid gfDNA values (ng/μL) increase with disease stage (T0-T2: 15.12; T3-T4: 30.75), and both are higher than the mean gfDNA values observed in non-neoplastic disease (10.81 ng/μL for N+PD and 10.10 ng/μL for

PN). These subject numbers in each diagnostic group accurately reflect real-world data from a tertiary cancer center.

(3) *“The prognosis evaluation is too simplistic, only considering staging factors, without taking into account other factors such as tumor pathology and the time from onset to tumor detection.”*

Histopathological analyses were performed throughout the study not only for the initial diagnosis of tissue biopsies, but also for the classification of Lauren’s subtypes, tumor staging, and the assessment of the presence and extent of immune cell infiltrates. Regarding the time of disease onset, this variable is inherently unknown—by definition—at the time of a diagnostic EGD. While the prognosis definition is indeed straightforward, we believe that a simple, cost-effective, and practical approach is advantageous for patients across diverse clinical settings and is more likely to be effectively integrated into routine EGD practice.

(4) *“The comparison between gfDNA and conventional pathological examination methods should be mentioned, reflecting advantages such as accuracy and patient comfort.”*

We wish to reinforce that EGD, along with conventional histopathology, remains the gold standard for gastric cancer evaluation. EGD under sedation is routinely performed for diagnosis, and the collection of gastric fluids for gfDNA evaluation does not affect patient comfort. Thus, while gfDNA analysis was evidently not intended as a diagnostic EGD and biopsy replacement, it may provide added prognostic value to this exam.

(5) *“There are many questions in the figures and tables. Please match the Title, Figure legends, Footnote, Alphabetic order, etc.”*

We are grateful for these comments and apologize for the clerical oversight. All figures, tables, titles and figure legends have now been double-checked.

(6) *“The overall logicity of the manuscript is not rigorous enough, with few discussion factors, and cannot represent the conclusions drawn.”*

We assume that the unusual wording remark regarding “overall logicity” pertains to the rationale and/or reasoning of this investigational study. Our working hypothesis was that during neoplastic disease progression, tumor cells continuously proliferate and, depending on various factors, attract immune cell infiltrates. Consequently, both tumor cells and immune cells (as well as tumor-derived DNA) are released into the fluids surrounding the tumor at its various locations, including blood, urine, saliva, gastric fluids, and others. Thus, increases in DNA levels within some of these fluids have been documented and are clinically meaningful. The concurrent observation of elevated gastric fluid gfDNA levels and immune cell infiltration supports the hypothesis that increased gfDNA—which may originate not only from tumor cells but also from immune cells—could be associated with better prognosis, as suggested by this study of a large real-world patient cohort.

In summary, we thank Reviewer #1 for his time and effort in a constructive critique of our work.

Reviewer #2 (Public review):

Summary:

“The authors investigated whether the total DNA concentration in gastric fluid (gfDNA), collected via routine esophagogastroduodenoscopy (EGD), could serve as a diagnostic and prognostic biomarker for gastric cancer. In a large patient cohort (initial n=1,056; analyzed n=941), they found that gfDNA levels were significantly higher in gastric cancer patients compared to non-cancer, gastritis, and precancerous lesion groups.

Unexpectedly, higher gfDNA concentrations were also significantly associated with better survival prognosis and positively correlated with immune cell infiltration. The authors proposed that gfDNA may reflect both tumor burden and immune activity, potentially serving as a cost-effective and convenient liquid biopsy tool to assist in gastric cancer diagnosis, staging, and follow-up."

Strengths:

"This study is supported by a robust sample size (n=941) with clear patient classification, enabling reliable statistical analysis. It employs a simple, low-threshold method for measuring total gfDNA, making it suitable for large-scale clinical use. Clinical confounders, including age, sex, BMI, gastric fluid pH, and PPI use, were systematically controlled. The findings demonstrate both diagnostic and prognostic value of gfDNA, as its concentration can help distinguish gastric cancer patients and correlates with tumor progression and survival. Additionally, preliminary mechanistic data reveal a significant association between elevated gfDNA levels and increased immune cell infiltration in tumors (p=0.001)."

Reviewer #2 has conceptually grasped the overall rationale of the study quite well, and we are grateful for their assessment and comprehensive summary of our findings.

Weaknesses:

(1) "The study has several notable weaknesses. The association between high gfDNA levels and better survival contradicts conventional expectations and raises concerns about the biological interpretation of the findings."

We agree that this would be the case if the gfDNA was derived solely from tumor cells. However, the findings presented here suggest that a fraction of this DNA would be indeed derived from infiltrating immune cells. The precise determination of the origin of this increased gfDNA remains to be achieved in future follow-up studies, and these are planned to be evaluated soon, by applying DNA- and RNA-sequencing methodologies and deconvolution analyses.

(2) "The diagnostic performance of gfDNA alone was only moderate, and the study did not explore potential improvements through combination with established biomarkers. Methodological limitations include a lack of control for pre-analytical variables, the absence of longitudinal data, and imbalanced group sizes, which may affect the robustness and generalizability of the results."

Reviewer #2 is correct that this investigational study was not designed to assess the diagnostic potential of gfDNA. Instead, its primary contribution is to provide useful prognostic information. In this regard, we have not yet explored combining gfDNA with other clinically well-established diagnostic biomarkers. We do acknowledge this current limitation as a logical follow-up that must be investigated in the near future.

Moreover, we collected a substantial number of pre-analytical variables within the limitations of a study involving over 1,000 subjects. Longitudinal samples and data were not analyzed here, as our aim was to evaluate prognostic value at diagnosis. Although the groups are imbalanced, this accurately reflects the real-world population of a large endoscopy center within a dedicated cancer facility. Subjects were invited to participate and enter the study before sedation for the diagnostic EGD procedure; thus, samples were collected prospectively from all consenting individuals.

Finally, to maintain a large, unbiased cohort, we did not attempt to balance the groups, allowing analysis of samples and data from all patients with compatible diagnoses (please see Results: Patient groups and diagnoses).

(3) “Additionally, key methodological details were insufficiently reported, and the ROC analysis lacked comprehensive performance metrics, limiting the study’s clinical applicability.”

We are grateful for this useful suggestion. In the current version, each ROC curve (Supplementary Figures 1A and 1B) now includes the top 10 gfDNA thresholds, along with their corresponding sensitivity and specificity values (please see Suppl. Table 1). The thresholds are ordered from-best-to-worst based on the classic Youden’s J statistic, as follows:

Youden Index = specificity + sensitivity – 1 [Youden WJ. Index for rating diagnostic tests. *Cancer* 3:32-35, 1950. PMID: 15405679]. We have made an effort to provide all the key methodological details requested, but we would be glad to add further information upon specific request.

Reviewer #1 (Recommendations for the authors):

The authors should pay attention to ensuring uniformity in the format of all cited references, such as the number of authors for each reference, the journal names, publication years, volume numbers, and page number formats, to the best extent possible.

Thank you for pointing this inconsistency. All cited references have now been revisited and adjusted properly. We apologize for this clerical oversight.

Reviewer #2 (Recommendations for the authors):

(1) “High gfDNA levels were surprisingly linked to better survival, which conflicts with the conventional understanding of cfDNA as a tumor burden marker. Was any qualitative analysis performed to distinguish DNA derived from immune cells versus tumor cells?”

Tumor-derived DNA is certainly present in gfDNA, as our group has unequivocally demonstrated in a previous publication [Pizzi M. P., et al. (2019) Identification of DNA mutations in gastric washes from gastric adenocarcinoma patients: Possible implications for liquid biopsies and patient follow-up *Int J Cancer* 145:1090–1097. DOI: 10.1002/ijc.32114]. However, in the present manuscript, our data suggest that gfDNA may also contain DNA derived from infiltrating immune cells. This may also be the case for other malignancies, and qualitative deconvolution studies could provide more informative information. To achieve this, DNA sequencing and RNA-Seq analyses may offer relevant evidence. Our study should be viewed as an original and preliminary analysis that may encourage such quantitative and qualitative studies in biofluids from cancer patients. Currently, this is a simple approach (which might be its essential beauty), but we hope to investigate this aspect further in future studies.

(2) “The ROC curve AUC was 0.66, indicating only moderate discrimination ability. Did the authors consider combining gfDNA with markers such as CEA or CA19-9 to improve diagnostic accuracy?”

This is indeed a logical idea, which shall certainly be explored in planned follow-up studies.

(3) “DNA concentration could be influenced by non-biological factors, including gastric fluid pH, sampling location, time delay, or freeze-thaw cycles. Were these operational variables assessed for their effect on data stability?”

We appreciate the rigor of the evaluation. Yes, information regarding gastric fluid pH was collected. All samples were collected from the stomach during EGD procedure. Samples were divided in aliquots and were thawed only once. This information is now provided in the updated manuscript text.

(4) *“This cross-sectional study lacks data on gfDNA changes over time, limiting conclusions on its utility for monitoring treatment response or predicting recurrence.”*

Again, temporal evaluation is another excellent point, and it will be the subject of future analyses. In this exploratory study, samples were collected at diagnosis, at a single point. We have not obtained serial samples, as participants received appropriate therapy soon following diagnosis.

(5) *The normal endoscopy group included only 10 patients, the precancerous lesion group 99 patients, while the gastritis group had 596 patients. Such uneven sample sizes may affect statistical reliability and generalizability. Has weighted analysis or optimized sampling been considered for future studies?”*

Yes, in future studies this analysis will be considered, probably by employing stratified random sampling with relevant patient attributes recorded.

(6) *“The SciScore was only 2 points, indicating that key methodological details such as inclusion/exclusion criteria, randomization, sex variables, and power calculation were not clearly described. It is recommended that these basic research elements be supplemented in the Methods section.”*

This was an exploratory research, the first of its kind, to evaluate prognostic potential of gfDNA in the context of gastric cancer. Patients were not included if they did not sign the informed consent or excluded if they withdrew after consenting. Other exclusion criteria included diagnoses of conditions such as previous gastrectomy or esophagectomy, or the presence of non-gastric malignancies. Randomization and power analyses were not applicable, as no prior data were available regarding gfDNA concentration values or its diagnostic/prognostic potential. All subjects, regardless of sex, were invited to participate without discrimination or selection.

(7) *“Although a ROC curve was provided in the supplementary materials (Supplementary Figure 1), only the curve and AUC value were shown without sensitivity, specificity, predictive values, or cutoff thresholds. The authors are advised to provide a full ROC performance assessment to strengthen the study's clinical relevance.”*

These data are now given alongside the ROC curves in the Supplementary Information section, specifically in Supplementary Figure 1 and in the newly added Supplementary Table 1.

We thank Reviewer #2 for an insightful and positive overall assessment of our work.

<https://doi.org/10.7554/eLife.107103.2.sa0>